BS-M301 — Probability and Statistics

Semester III • 3L:0T:3P • 3 Credits

Module 2 Exam Notes: Continuous Probability Distributions

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Continuous Random Variables, PDF and CDF

A random variable is classified as **continuous** if its range consists of an entire interval or a union of intervals on the real line $\mathbb{R}$ (such as voltage, time, temperature, or noise amplitude). Because an interval contains uncountably infinitely many points, the probability that a continuous random variable assumes any exact single real value is strictly zero: $P(X = c) = 0$ for every $c \in \mathbb{R}$.

Definition: Probability Density Function (PDF)

A function $f_X : \mathbb{R} \to \mathbb{R}$ is called the **Probability Density Function (PDF)** of a continuous random variable X if for any interval $[a, b] \subseteq \mathbb{R}$:

$$P(a \leq X \leq b) = \int_a^b f_X(x)\, dx \tag{1}$$

Every valid PDF must strictly satisfy two axiomatic criteria:

$$\textbf{Non-negativity:} \quad f_X(x) \geq 0, \quad \forall x \in \mathbb{R} \tag{2}$$

$$\textbf{Total Normalization:} \quad \int_{-\infty}^{\infty} f_X(x)\, dx = 1 \tag{3}$$

Crucial Insight: The value $f_X(x)$ represents probability *density* per unit length, not probability itself! Consequently, $f_X(x)$ can exceed 1 over narrow intervals, provided the total area under the curve equals unity.

1.1 Cumulative Distribution Function (CDF)

The **Cumulative Distribution Function (CDF)** of a continuous random variable X is defined as the area under the PDF curve from $-\infty$ up to threshold x:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\, dt \tag{4}$$

By the Fundamental Theorem of Calculus, at every point where $f_X(x)$ is continuous, the PDF is the exact first derivative of the CDF:

$$f_X(x) = \frac{d}{dx} F_X(x) \tag{5}$$

Key Properties of Continuous CDF and Interval Probabilities

1. **Continuous and Smooth:** Unlike discrete CDFs (which have jump discontinuities), the CDF $F_X(x)$ of a continuous random variable is everywhere continuous.
2. **Boundary Limits:** $\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to +\infty} F_X(x) = 1$.
3. **Boundary Invariance of Interval Probabilities:** Since single-point probabilities are

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zero ($P(X = a) = P(X = b) = 0$):

$$P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b)$$
$$= F_X(b) - F_X(a) = \int_a^b f_X(x)\,dx \tag{6}$$

4. **Tail (Survival) Function:** $R_X(x) = P(X > x) = 1 - F_X(x) = \int_x^\infty f_X(t)\,dt$.

1.2 Expectation, Variance and Moment Generating Function (MGF)

- **Mathematical Expectation (Mean):**

$$\mu = E[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx \quad \text{(provided } \int_{-\infty}^{\infty} |x| f_X(x)dx < \infty) \tag{7}$$

- **Variance (σ^2):**

$$\text{Var}(X) = \sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f_X(x)\,dx = E[X^2] - (E[X])^2 \tag{8}$$

- **Moment Generating Function (MGF):** The MGF $M_X(t)$ is defined as the Laplace-type transform:

$$M_X(t) = E\left[e^{tX}\right] = \int_{-\infty}^{\infty} e^{tx} f_X(x)\,dx \tag{9}$$

  provided the integral converges in an open neighborhood around $t = 0$ ($|t| < h$).

- **Generation of Moments from MGF:** Expanding $e^{tX} = \sum_{r=0}^{\infty} \frac{t^r X^r}{r!}$:

$$M_X(t) = \sum_{r=0}^{\infty} \frac{t^r}{r!} E[X^r] = 1 + \mu_1' t + \frac{\mu_2' t^2}{2!} + \frac{\mu_3' t^3}{3!} + \cdots \tag{10}$$

  Differentiating r times and setting $t = 0$:

$$\mu_r' = E[X^r] = \left.\frac{d^r M_X(t)}{dt^r}\right|_{t=0} \tag{11}$$

2. The Normal (Gaussian) Distribution

The **Normal distribution** (discovered by Carl Friedrich Gauss) is the central pillar of engineering probability, signal analysis, and statistics. By the Central Limit Theorem, the sum of a large number of independent random disturbances converges to a normal distribution, regardless of the underlying distribution of the individual components.

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Definition: Normal Distribution $\mathcal{N}(\mu, \sigma^2)$

A continuous random variable X has a **Normal distribution** with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$ (written $X \sim \mathcal{N}(\mu, \sigma^2)$) if its PDF is:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad -\infty < x < \infty \tag{12}$$

2.1 Mathematical Verification of the Normalizing Constant

To verify $\int_{-\infty}^{\infty} f_X(x)dx = 1$, let $I = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)} dx$. Substitute the standard normal transform $z = \frac{x-\mu}{\sigma} \implies dx = \sigma dz$:

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2}\, dz \tag{13}$$

Let $J = \int_{-\infty}^{\infty} e^{-z^2/2} dz$. Converting J^2 to polar coordinates ($x = r\cos\theta, y = r\sin\theta, dxdy = rdrd\theta$):

$$J^2 = \int_{-\infty}^{\infty} e^{-x^2/2} dx \int_{-\infty}^{\infty} e^{-y^2/2} dy = \int_0^{2\pi} d\theta \int_0^{\infty} re^{-r^2/2}\, dr = 2\pi \left[-e^{-r^2/2}\right]_0^{\infty} = 2\pi(1) = 2\pi \tag{14}$$

Hence $J = \sqrt{2\pi}$, confirming $I = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1$.

Derivation: MGF and Moments of the Normal Distribution

For $X \sim \mathcal{N}(\mu, \sigma^2)$, the MGF is:

$$M_X(t) = E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}\, dx \tag{15}$$

Substituting $z = \frac{x-\mu}{\sigma} \implies x = \mu + \sigma z$ and $dx = \sigma dz$:

$$M_X(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t(\mu+\sigma z)} e^{-z^2/2}\, dz = e^{\mu t} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}(z^2 - 2\sigma t z)\right) dz \tag{16}$$

Completing the square in the exponent: $z^2 - 2\sigma tz = (z-\sigma t)^2 - \sigma^2 t^2$:

$$M_X(t) = e^{\mu t} e^{\sigma^2 t^2/2} \cdot \underbrace{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(z-\sigma t)^2/2}\, dz}_{=1 \text{ (integral of standard normal density)}} = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right) \tag{17}$$

Moments via Derivatives of MGF:

$$M_X'(t) = (\mu + \sigma^2 t) e^{\mu t + \frac{1}{2}\sigma^2 t^2} \implies E[X] = M_X'(0) = \mu \tag{18}$$

$$M_X''(t) = \left[\sigma^2 + (\mu + \sigma^2 t)^2\right] e^{\mu t + \frac{1}{2}\sigma^2 t^2} \implies E[X^2] = M_X''(0) = \sigma^2 + \mu^2 \tag{19}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = (\sigma^2 + \mu^2) - \mu^2 = \sigma^2 \tag{20}$$

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2.2 Symmetry, Odd Moments and Inflection Points

1. **Symmetry about μ:** The bell curve is symmetric about the vertical line $x = \mu$: $f_X(\mu - x) = f_X(\mu + x)$. Consequently:

$$\text{Mean} = \text{Median} = \text{Mode} = \mu \tag{21}$$

2. **Vanishing of All Odd Central Moments:** By anti-symmetry of odd powers about zero:

$$\mu_{2k+1} = E[(X-\mu)^{2k+1}] = \int_{-\infty}^{\infty} (x-\mu)^{2k+1} f_X(x)\, dx = 0 \quad \forall k \in \mathbb{N}_0 \tag{22}$$

Specifically $\mu_1 = 0$ and $\mu_3 = 0$. Hence the skewness coefficient is identically zero: $\beta_1 = \gamma_1 = 0$.

3. **Even Central Moments and Kurtosis:**

$$\mu_{2k} = 1 \cdot 3 \cdot 5 \cdots (2k-1)\, \sigma^{2k} = \frac{(2k)!}{2^k k!} \sigma^{2k} \implies \mu_4 = 3\sigma^4 \tag{23}$$

The coefficient of kurtosis is: $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{3\sigma^4}{(\sigma^2)^2} = 3$ (Mesokurtic benchmark).

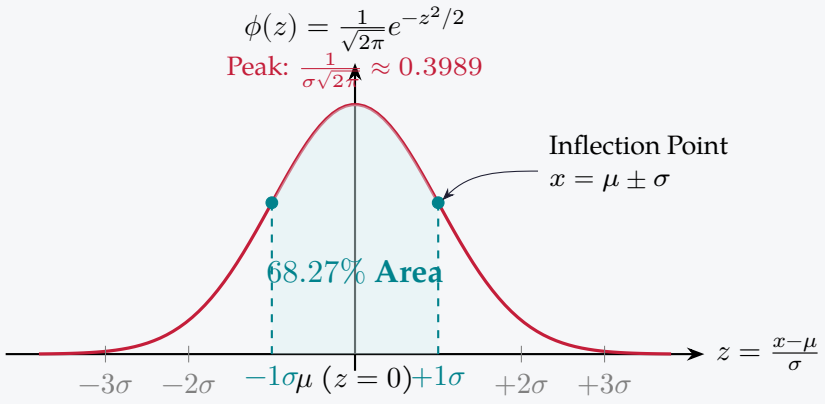
4. **Points of Inflection:** Finding where the concavity changes ($\frac{d^2 f_X}{dx^2} = 0$):

$$f_X'(x) = -\frac{x-\mu}{\sigma^2} f_X(x) \tag{24}$$

$$f_X''(x) = \left[\left(\frac{x-\mu}{\sigma^2} \right)^2 - \frac{1}{\sigma^2} \right] f_X(x) = 0 \implies (x-\mu)^2 = \sigma^2 \implies x = \mu \pm \sigma \tag{25}$$

The curve shifts from concave-downward to concave-upward at exactly $x = \mu \pm \sigma$.

TikZ Schematic: Standard Normal Gaussian Curve with Inflection Points and Area Rules



2.3 Standard Normal Distribution and $\Phi(z)$ Function

When $\mu = 0$ and $\sigma = 1$, the variable is denoted $Z \sim \mathcal{N}(0,1)$ with standard normal PDF $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and CDF:

$$\Phi(z) = P(Z \le z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-u^2/2}\, du \tag{26}$$

By symmetry:

$$\Phi(-z) = 1 - \Phi(z) \qquad \text{and} \qquad P(a \le X \le b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right) \tag{27}$$

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3. The Exponential Distribution

The **Exponential distribution** models the continuous time elapsed between independent random events occurring at a constant average rate $\lambda > 0$ (such as service times, component lifetimes, and inter-arrival times in queueing systems).

Definition: Exponential Distribution Exp(□)

A continuous random variable X has an **Exponential distribution** with rate parameter $\lambda > 0$ ($X \sim \text{Exp}(\lambda)$) if its PDF is:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases} \tag{28}$$

3.1 Derivation of CDF, Mean, Variance and MGF

1. **Cumulative Distribution Function (CDF):** For $x \geq 0$:

$$F_X(x) = \int_0^x \lambda e^{-\lambda t}\, dt = \left[-e^{-\lambda t}\right]_0^x = 1 - e^{-\lambda x} \tag{29}$$

The survival (tail) probability is:

$$P(X > x) = 1 - F_X(x) = e^{-\lambda x} \tag{30}$$

2. **Moment Generating Function (MGF):** For $t < \lambda$:

$$M_X(t) = \int_0^\infty e^{tx} \lambda e^{-\lambda x}\, dx = \lambda \int_0^\infty e^{-(\lambda - t)x}\, dx = \frac{\lambda}{\lambda - t} = \left(1 - \frac{t}{\lambda}\right)^{-1} \tag{31}$$

Expanding in powers of t/λ:

$$M_X(t) = 1 + \frac{t}{\lambda} + \frac{t^2}{\lambda^2} + \frac{t^3}{\lambda^3} + \cdots = 1 + \left(\frac{1}{\lambda}\right) t + \left(\frac{2}{\lambda^2}\right) \frac{t^2}{2!} + \cdots \tag{32}$$

3. **Mean and Variance:**

$$\mu = E[X] = \frac{1}{\lambda}, \qquad E[X^2] = \frac{2}{\lambda^2} \implies \text{Var}(X) = E[X^2] - (E[X])^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2} \tag{33}$$

Standard deviation is $\sigma = \frac{1}{\lambda}$. Notice that for the Exponential distribution: $\sigma = \mu$.

3.2 The Memoryless Property (Markov Property)

Theorem: Memoryless Property of the Exponential Distribution

Let $X \sim \text{Exp}(\lambda)$. For any non-negative time increments $s, t \geq 0$:

$$P(X > s + t \mid X > s) = P(X > t) \tag{34}$$

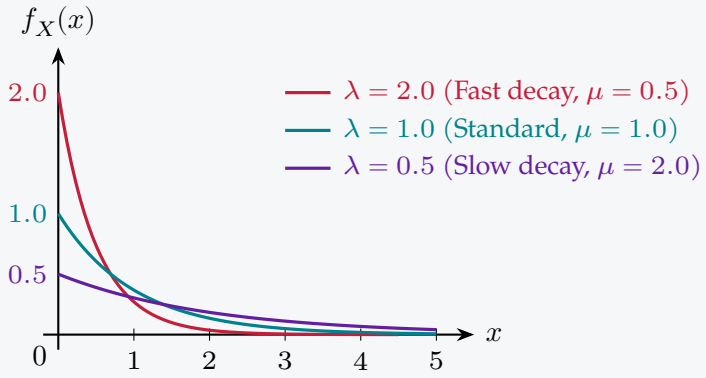
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Proof: Using the conditional probability definition and $P(X > x) = e^{-\lambda x}$:

$$P(X > s+t \mid X > s) = \frac{P(\{X > s+t\} \cap \{X > s\})}{P(X > s)}$$
$$= \frac{P(X > s+t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = \frac{e^{-\lambda s}e^{-\lambda t}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t) \tag{35}$$

Physical Interpretation: If a semiconductor component has survived s hours without failing, the probability that it survives an additional t hours is identical to the probability that a brand-new component survives t hours. The component suffers no wear-out aging over time!

TikZ Plot: Exponential Probability Density Functions for Different Rates λ



4. The Gamma Distribution and Its Properties

The **Gamma distribution** is a versatile two-parameter continuous distribution that extends the exponential distribution to model the waiting time until the occurrence of α successive Poisson events, as well as lifetime data exhibiting wear-out characteristics.

Definition: The Gamma Function $\Gamma(\alpha)$

The complete **Eulerian Gamma Function** is defined for all $\alpha > 0$ by the improper definite integral:

$$\Gamma(\alpha) = \int_0^\infty u^{\alpha-1}e^{-u}\,du \tag{36}$$

Fundamental Properties of $\Gamma(\alpha)$:

1. **Recurrence Relation:** $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ (integrating by parts).
2. **Integer Factorial:** For any positive integer $n \in \mathbb{N}$: $\Gamma(n) = (n-1)!$ with $\Gamma(1) = 0! = 1$.
3. **Half-Integer Values:** $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, from which $\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\sqrt{\pi}$, $\Gamma\left(\frac{5}{2}\right) = \frac{3}{4}\sqrt{\pi}$.

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Two-Parameter Gamma Density Gamma(□, □)

A random variable X follows a **Gamma distribution** with shape parameter $\alpha > 0$ and rate parameter $\beta > 0$ (scale parameter $\theta = 1/\beta$) if its PDF is:

$$f_X(x) = \begin{cases} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, & x > 0 \\ 0, & x \leq 0 \end{cases} \tag{37}$$

4.1 Derivation of Mean, Variance and MGF of Gamma Distribution

Rigorous Derivation: Moments of Gamma Distribution

Using the MGF definition for $t < \beta$:

$$M_X(t) = \int_0^\infty e^{tx} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}\, dx = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-(\beta-t)x}\, dx \tag{38}$$

Substitute $u = (\beta - t)x \implies x = \frac{u}{\beta-t}$ and $dx = \frac{du}{\beta-t}$:

$$M_X(t) = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty \left(\frac{u}{\beta-t}\right)^{\alpha-1} e^{-u} \frac{du}{\beta-t} = \frac{\beta^\alpha}{(\beta-t)^\alpha} \cdot \underbrace{\frac{1}{\Gamma(\alpha)} \int_0^\infty u^{\alpha-1} e^{-u} du}_{=1} = \left(1 - \frac{t}{\beta}\right)^{-\alpha} \tag{39}$$

Expanding $(1 - t/\beta)^{-\alpha}$ using the generalized binomial theorem:

$$\begin{aligned} M_X(t) &= 1 + (-\alpha)\left(-\frac{t}{\beta}\right) + \frac{(-\alpha)(-\alpha-1)}{2!}\left(-\frac{t}{\beta}\right)^2 + \cdots \\ &= 1 + \left(\frac{\alpha}{\beta}\right) t + \frac{\alpha(\alpha+1)}{\beta^2} \frac{t^2}{2!} + \cdots \end{aligned} \tag{40}$$

Equating coefficients of t and $t^2/2!$:

$$E[X] = \mu = \frac{\alpha}{\beta} \tag{41}$$

$$E[X^2] = \frac{\alpha(\alpha+1)}{\beta^2} \implies \text{Var}(X) = E[X^2] - (E[X])^2 = \frac{\alpha^2 + \alpha}{\beta^2} - \frac{\alpha^2}{\beta^2} = \frac{\alpha}{\beta^2} \tag{42}$$

4.2 Special Cases and Reproductive Properties of Gamma Distribution

1. **Exponential Distribution Special Case:** When shape parameter $\alpha = 1$:

$$f_X(x) = \frac{\beta^1}{\Gamma(1)} x^{1-1} e^{-\beta x} = \beta e^{-\beta x} \equiv \text{Exp}(\beta) \tag{43}$$

2. **Erlang Distribution:** When shape parameter $\alpha = k$ is a positive integer ($k \in \mathbb{N}$), the distribution represents the sum of k independent and identically distributed $\text{Exp}(\beta)$ variables.

3. **Chi-Square (χ^2) Distribution Connection:** Substituting $\alpha = \frac{n}{2}$ and $\beta = \frac{1}{2}$ into the

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Gamma PDF yields:

$$f_X(x) = \frac{(1/2)^{n/2}}{\Gamma(n/2)} x^{n/2-1} e^{-x/2} \equiv \chi^2(n) \quad \text{(Chi-square with } n \text{ degrees of freedom)} \tag{44}$$

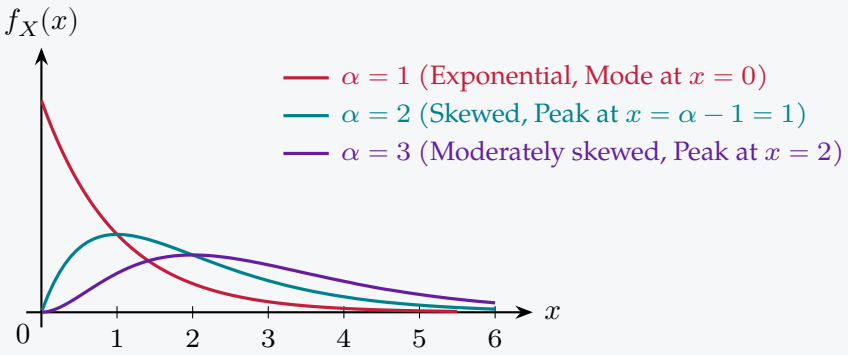
4. **Reproductive (Additive) Property:** If $X_1 \sim \text{Gamma}(\alpha_1, \beta)$ and $X_2 \sim \text{Gamma}(\alpha_2, \beta)$ are independent random variables sharing the *same* rate parameter β, then:

$$M_{X_1+X_2}(t) = M_{X_1}(t) \cdot M_{X_2}(t) = \left(1 - \frac{t}{\beta}\right)^{-\alpha_1} \cdot \left(1 - \frac{t}{\beta}\right)^{-\alpha_2} = \left(1 - \frac{t}{\beta}\right)^{-(\alpha_1+\alpha_2)} \tag{45}$$

By the uniqueness theorem of MGFs:

$$X_1 + X_2 \sim \text{Gamma}(\alpha_1 + \alpha_2, \beta) \tag{46}$$

TikZ Plot: Gamma Family PDFs for $\beta = 1.0$ and Varying Shape Parameter α



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Quick Revision — Module 2

Continuous Probability Distributions | BS-M301

Key Formulas and Continuous Distribution Densities at a Glance

Continuous PDF Conditions	$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x)dx = 1, \quad P(a \leq X \leq b) = \int_a^b f_X(x)dx$
CDF to PDF Relationship	$F_X(x) = \int_{-\infty}^{x} f_X(t)dt, \quad f_X(x) = \frac{d}{dx}F_X(x), \quad P(X = c) = 0$
Continuous Expectation	$E[X] = \mu = \int_{-\infty}^{\infty} x f_X(x)dx, \quad \text{Var}(X) = E[X^2] - \mu^2$
MGF Formula	$M_X(t) = E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f_X(x)dx, \quad \mu_r' = \frac{d^r M_X(t)}{dt^r}\Big\|_{t=0}$
Normal PDF $\mathcal{N}(\mu, \sigma^2)$	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)$
Standard Normal Z	$Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1), \quad \phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad \Phi(-z) = 1 - \Phi(z)$
Normal Moments	All odd $\mu_{2k+1} = 0, \quad \mu_2 = \sigma^2, \quad \mu_4 = 3\sigma^4, \quad \beta_1 = 0, \ \beta_2 = 3$
Exponential PDF Exp(λ)	$f(x) = \lambda e^{-\lambda x} \ (x \geq 0), \quad F(x) = 1 - e^{-\lambda x}, \quad \mu = \frac{1}{\lambda}, \ \sigma^2 = \frac{1}{\lambda^2}$
Memoryless Property	$P(X > s + t \mid X > s) = P(X > t)$ (Unique to Exponential)
Gamma Function	$\Gamma(\alpha) = \int_0^{\infty} u^{\alpha-1} e^{-u} du, \quad \Gamma(\alpha+1) = \alpha\Gamma(\alpha), \quad \Gamma(1/2) = \sqrt{\pi}$
Gamma PDF Gamma(α, β)	$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad \mu = \frac{\alpha}{\beta}, \quad \sigma^2 = \frac{\alpha}{\beta^2}, \quad M_X(t) = \left(1 - \frac{t}{\beta}\right)^{-\alpha}$

One-Line Definitions

- **Continuous Random Variable:** Variable taking any real value in an interval, with single-point probability $P(X = c) = 0$.
- **Probability Density Function (PDF):** Non-negative derivative of the CDF whose integral over an interval gives probability.
- **Normal Distribution:** Symmetrical mesokurtic bell curve fully governed by location μ and scale σ.
- **Inflection Points:** Points where curvature flips concavity, situated at precisely $x = \mu \pm \sigma$ for a normal curve.
- **Standard Score (Z):** Dimensionless measure $Z = (X - \mu)/\sigma$ expressing deviation from mean in units of σ.
- **Exponential Distribution:** Only continuous distribution possessing memorylessness, modelling Poisson inter-arrival times.
- **Memoryless Property:** Independence of future remaining lifetime from past survival duration: $P(X > s + t \mid X > s) = P(X > t)$.
- **Gamma Function ($\Gamma(\alpha)$):** Generalized factorial integral extending $n!$ to all real positive numbers ($\Gamma(n) = (n-1)!$).
- **Erlang Distribution:** Gamma distribution with integer shape parameter $\alpha = k$, representing sum of k independent exponentials.
- **Reproductive Property:** Closure under convolution: sum of independent Gammas with same β yields another Gamma.

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Frequently Asked / Viva Questions (Part 1)

Q1. Why is the probability of a continuous random variable taking an exact value zero ($P(X = c) = 0$)?

By definition, $P(X = c) = \int_c^c f_X(x)\,dx = 0$. Since an interval contains uncountably infinitely many points, the probability mass allocated to any isolated geometric point vanishes, even though the event is not impossible in reality.

Q2. Can a probability density function exceed unity ($f_X(x) > 1$)? Explain with an example.

Yes! Unlike discrete PMFs where $p(x) \leq 1$, a continuous PDF represents probability *density* per unit width, not probability. For example, for $X \sim U[0, 0.2]$, $f_X(x) = 1/(0.2 - 0) = 5.0 > 1$, while the total integral $\int_0^{0.2} 5.0\,dx = 1.0$ is fully preserved.

Q3. Derive the Moment Generating Function of the standard normal distribution $Z \sim \mathcal{N}(0, 1)$.

$M_Z(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz} e^{-z^2/2} dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(z^2-2tz)/2} dz$. Completing the square: $z^2 - 2tz = (z - t)^2 - t^2$. Factoring out $e^{t^2/2}$: $M_Z(t) = e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(z-t)^2/2} dz = e^{t^2/2} \cdot 1 = e^{t^2/2}$.

Q4. Prove that all odd central moments of a Normal distribution are identically zero.

$\mu_{2k+1} = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x - \mu)^{2k+1} e^{-(x-\mu)^2/(2\sigma^2)} dx$. Substituting $u = x - \mu$: the integrand $u^{2k+1} e^{-u^2/(2\sigma^2)}$ is an odd function of u ($g(-u) = -g(u)$) integrated over symmetric limits $(-\infty, \infty)$, which integrates to zero.

Q5. Where do the inflection points of a normal probability curve occur?

Equating the second derivative to zero: $f''(x) = \left[\frac{(x-\mu)^2}{\sigma^4} - \frac{1}{\sigma^2}\right] f(x) = 0 \implies (x - \mu)^2 = \sigma^2 \implies x = \mu \pm \sigma$. The curve transitions from concave downwards to concave upwards at exactly one standard deviation on either side of the mean.

Q6. State and prove the memoryless property of the Exponential distribution.

$P(X > s + t \mid X > s) = \frac{P(X>s+t)}{P(X>s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)$. The conditional probability that a component lasts t more units given it has lasted s units is independent of s.

Q7. What is the physical connection between the Poisson process and the Exponential distribution?

In a Poisson process with rate λ, the number of event arrivals in interval t follows Poisson(λt). The waiting time T until the first arrival exceeds t if and only if zero arrivals occur in $[0, t]$: $P(T > t) = P(N(t) = 0) = \frac{(\lambda t)^0 e^{-\lambda t}}{0!} = e^{-\lambda t}$. Thus, the inter-arrival time T is exponentially distributed with parameter λ.

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Frequently Asked / Viva Questions (Part 2)

Q8. Prove that $\Gamma(1/2) = \sqrt{\pi}$.

$\Gamma(1/2) = \int_0^\infty u^{-1/2}e^{-u}du$. Substituting $u = z^2/2 \implies du = zdz$: $\Gamma(1/2) = \int_0^\infty \frac{\sqrt{2}}{z}e^{-z^2/2}zdz = \sqrt{2}\int_0^\infty e^{-z^2/2}dz$. By symmetry: $\int_0^\infty e^{-z^2/2}dz = \frac{1}{2}\int_{-\infty}^\infty e^{-z^2/2}dz = \frac{\sqrt{2\pi}}{2} = \sqrt{\frac{\pi}{2}}$. Multiplying by $\sqrt{2}$ yields $\Gamma(1/2) = \sqrt{\pi}$.

Q9. How is the Chi-Square (χ^2) distribution obtained as a special case of the Gamma distribution?

Setting shape parameter $\alpha = n/2$ and rate parameter $\beta = 1/2$ in the Gamma PDF yields $f(x) = \frac{(1/2)^{n/2}}{\Gamma(n/2)}x^{n/2-1}e^{-x/2}$, which is the standard definition of the Chi-Square distribution with n degrees of freedom.

Q10. State the reproductive property of independent Normal random variables.

If $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, any linear combination $Y = aX_1 + bX_2$ is also normally distributed: $Y \sim \mathcal{N}(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$. This follows directly from multiplying their MGFs.

Summary Matrix: Comprehensive Comparison of Continuous Distributions

Distribution	Probability Density Function (PDF)	Mean ($E[X]$)	Variance (σ^2)	Key Engineering Application
Uniform $U[a,b]$	$f(x) = \frac{1}{b-a}, \quad x \in [a,b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	Quantization noise in ADC systems.
Normal $\mathcal{N}(\mu,\sigma^2)$	$f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2	Thermal Gaussian noise; Central Limit Theorem.
Exponential **Exp**(λ)	$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	Lifetime of electronic parts; queueing theory.
Gamma (α,β)	$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}, \ x > 0$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	Waiting time for α events; reliability engineering.
Chi-Square $\chi^2(n)$	$f(x) = \frac{(1/2)^{n/2}}{\Gamma(n/2)}x^{n/2-1}e^{-x/2}$	n	$2n$	Goodness-of-fit testing; sample variance distribution.

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